

Le Nguyen Gia Hung

AI/ML Researcher

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Education

FPT University

Bachelor of Engineering in Artificial Intelligence

GPA: 8.0/10.0

Ho Chi Minh City, Vietnam

Expected Dec 2027

- **Honors:** Certificates of Merit – Honorable Student of Trimester (Semesters 3, 4 & 5).
- **Relevant Coursework:** Data Structures and Algorithms, Linear Algebra, Introduction to Artificial Intelligence, Deep Learning, Statistical Analysis.

Publications

2025 **Gia-Hung Nguyen Le et al.**, “Proactive Air Quality Forecasting and Health Alert System for Melbourne.” *Australasian Joint Conference on Artificial Intelligence (AJCAI 2025)*. Springer, Lecture Notes in Artificial Intelligence (LNAI). Q2 (SJR 2024).
DOI: 10.1007/978-981-95-4969-6_34

Research & Technical Experience

Proactive Air Quality Forecasting System (AJCAI 2025)

Lead Researcher & Author

Academic Research

Jan 2025 – Present

- Architected and implemented **SmokeNet**, a novel Transformer-based architecture for time-series forecasting of $PM_{2.5}$.
- Achieved **MAE of 0.7470** and R^2 of **0.9545**, reducing error by **35.9%** compared to XGBoost baselines on 2024 test data.
- Demonstrated model robustness by outperforming baselines by **57.7%** during the severe May 2024 smoke event backtest.
- Engineered an end-to-end pipeline processing 4+ years of OpenWeather data with GPU-accelerated chronological cross-validation.
- Integrated Explainable AI (SHAP, Integrated Gradients) to provide interpretable health warnings for public safety dashboards.

End-to-End Vietnamese Speech Recognition System

Co-Researcher

Academic Research

In Progress

- Developing a parameter-efficient (5.4M) end-to-end ASR model using a **4-layer CNN frontend** and **3-layer Residual BiLSTM encoder**.
- Curated and trained on a massive **745-hour Vietnamese speech corpus**, optimizing with Connectionist Temporal Classification (CTC) loss.
- Achieved a **Word Error Rate (WER) of 33.09%** and **Character Error Rate (CER) of 15.28%** on diverse test sets.
- Benchmarked performance against foundation models (PhoWhisper) to evaluate trade-offs between computational efficiency and accuracy.

Conformer-GAT-Fusion for Speech Emotion Recognition

Research Lead

Academic Research

In Progress

- Developing a hybrid **Conformer-GAT** architecture to fuse sequential audio features with graph-based textual interactions for multimodal emotion recognition.
- Leveraging Graph Attention Networks (GAT) to dynamically weight cross-modal dependencies between speech and text modalities.

- Benchmarking performance against state-of-the-art models on standard datasets (IEMOCAP, RAVDESS) with a focus on model interpretability.

Personal Protective Equipment (PPE) Detection System

Developer

Course Project

Sep 2024 – Dec 2024

- Built a real-time object detection system using **YOLOv8** to identify 9 classes of personal protective equipment (helmets, vests, gloves).
- Developed a safety logic engine to automatically classify workers as "Compliance Safe" or "Unsafe" based on detected equipment combinations.
- Deployed the model via a **Streamlit** web interface featuring live webcam inference and automated violation reporting.

Leadership

SpeedyLabX - Student Research Group

Founder & Lead Researcher

FPT University

2025 – Present

- Established and directed a research team of 10 undergraduates focused on applied AI solutions for environmental challenges.
- Spearheaded the Melbourne Air Quality project, managing timeline and deliverables, resulting in the group's first accepted publication at a top-tier regional conference (AJCAI).

Certifications

- **Building RAG Agents with LLMs** – NVIDIA
- **IBM Full Stack Software Developer Professional Certificate** – IBM
- **Data Science Fundamentals with Python and SQL** – IBM

Skills

- **Programming Languages:** Python, SQL (PostgreSQL), C++.
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, OpenCV.
- **Data & Tools:** Pandas, NumPy, Matplotlib, Git, Docker, Streamlit, Weights & Biases (W&B), Linux.
- **Technical Writing:** Authored "My First Encounter with AI" (Personal Tech Blog).
- **Areas of Interest:** Time-Series Forecasting, Transformer Architectures, Multimodal Learning, Explainable AI (XAI).